DANIEL NG

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Education

University of British Columbia — **Expected Graduation: May 2028**
*BASc in Electrical Engineering (**Co-op**, available May 2026)* — *Vancouver, BC*

Experience

Norsat — **June 2025 – August 2025**
Hardware Test Intern — *Richmond, BC*

- Characterized Ku-Band Block-Up Converters using Vector Network Analyzers, Signal Analyzers, oscilloscopes, and multimeters to ensure 100% compliance with hardware specifications.
- Optimized RF power module stability by precisely sweeping DAC control codes, significantly reducing variance in output power prior to final acceptance testing.
- Assembled and troubleshot BUCs and LNBs at the PCBA level, performing microscopic SMT and THT soldering and inspection to maintain quality standards for MMIC chips.
- Ensured 100% yield across a production run of 50+ BUC units by identifying DC bias instabilities through schematic cross-referencing.

UBC Formula Design Team — **September 2025 – Present**
Electrical & Firmware: Project Lead — *Vancouver, BC*
GitHub

- Lead end-to-end development of a laser receiver, incorporating optomechanical shrouding with a custom 3.3 V CMOS analog front end and 2.4 GHz ESP-NOW communication between master-slave ESP32-S3 MCUs to calculate speed and lap times.
- Characterize baseline photodiode thermal dark current and TIA voltage noise to calibrate an adjustable hysteresis threshold via potentiometer, targeting a 20 dB SNR and BER $< 10^{-7}$ to ensure high-speed data integrity.

Projects

Oven Reflow Controller | *A51 Assembly, CV-8052, NEC Protocol, I^2C* — **January 2026**
GitHub

- Architected a bare-metal thermal control system on a CV-8052 soft-core (DE10-Lite FPGA), implementing a deterministic FSM with adaptive PWM-based closed-loop control.
- Integrated I^2C peripherals, NEC IR protocol decoding (timing-validated bitstream parsing), and UART telemetry within a non-blocking interrupt-driven architecture.
- Developed a Python-based calibration and validation framework (NumPy/Matplotlib/pySerial) to stream live ADC data, perform thermocouple linearization and cold-junction compensation, cross-validate against laboratory instrumentation, and inject corrected temperature feedback to the controller — achieving ±3°C closed-loop accuracy across 25–240°C.

FPGA VGA Driver | *Embedded C, DE10-Lite FPGA, VGA* — **November 2025**
GitHub

- Developed a bare-metal VGA driver using Memory-Mapped I/O (MMIO) to interface with an FPGA pixel buffer, implementing efficient 2D-to-1D address mapping via bitwise shifting.
- Engineered a pixel-processing pipeline to convert 24-bit RGB values to the RGB565 color format, using bit-masking and packing to optimize memory bandwidth.
- Engineered an autonomous robot opponent with predictive collision logic, performing real-time memory reads of the VGA buffer to scan for obstacles and update movement vectors.

Skills

Digital Design/HDL/Low-Level: SystemVerilog, RISC-V, A51 Assembly, FPGA
Embedded Systems: CAN, USB, SPI, UART, I^2C, ESP32, 8051-family MCUs, C Embedded Firmware, Bare Metal
Design Tools: Quartus Prime, Questa/ModelSim, Altium Designer, CPUlator
Hardware & Lab: VNA, Signal Analyzer, PCB Design and Debugging, SMT/THT Soldering, Oscilloscope, Multimeter

Hobbies/Interests

Competitive Badminton • Chess • Reverse engineering hardware • Parallel Computing • Graphics Pipeline Architecture